

Appendix

1 Detailed Category Mapping for VidInc-CrossDomain

Table 1 provides the exhaustive mapping between the **Unified Category** and the original labels in ActivityNet, UCF101, and FCVID.

Table 1: Category mapping for VidInc-CrossDomain.

Unified Category	ActivityNet	UCF101	FCVID
Archery	Archery	Archery	archery
Bowling	Playing ten pins	Bowling	bowling
Fencing	Doing fencing	Fencing	fencing
Rope Skipping	Rope skipping	JumpRope	ropeSkipping
Table Tennis	Ping-pong	TableTennisShot	tableTennis
Tai Chi	Tai chi	TaiChi	taiChiChuan
Boxing	Doing kickboxing	BoxingPunchingBag, BoxingSpeedBag	boxing, punchingBag-Workout
Cycling & Biking	Assembling bicycle, BMX, Fixing bicycle	Biking	assemblingABike, bikeTricks, biking
Extreme Aerial Sports	Bungee jumping, Powerbocking	SkyDiving	bungeeJumping, skydiving
Keyboard Instrument Performance	Playing piano	PlayingPiano	pianoPerformance
Skateboarding & Skating	Longboarding, Skateboarding	SkateBoarding	rollerSkating, skateboarding
Animal Riding & Competition	Camel ride, Calf roping, Horseback riding, Playing polo	HorseRiding, HorseRace	camel, horseRiding
Dance	Ballet, Belly dance, Breakdancing, Cumbia, Tango, Zumba	SalsaSpin, IceDancing	groupDance, socialDance, soloDance, weddingDance
Diving	Plataform diving, Springboard diving	Diving, CliffDiving	diving
Football / Soccer	Beach soccer, Futsal, Playing kickball	SoccerJuggling, SoccerPenalty	soccerAmateur, soccerProfessional
Frisbee Sports	Disc dog	FrisbeeCatch	playingFrisbeeWithDog, playingFrisbeeWithPeople
Paddle Water Sports	Canoeing, Kayaking, Rafting, River tubing	Kayaking, Rafting, Rowing	boating, rafting, river, rowing
Painting & Writing	Painting, Painting fence, Painting furniture	WritingOnBoard	doingGraffiti, painting, sprayPainting
Percussion Instrument Performance	Playing drums, Playing congas	Drumming, PlayingTabla	beatbox, rockBandPerformance
Rock Climbing & Rope Climbing	Rock climbing	RockClimbingIndoor, RopeClimbing	rockClimbing

Continued on next page

Table 1 – continued from previous page

Unified Category	ActivityNet	UCF101	FCVID
Basketball	Layup drill in basketball	Basketball, BasketballDunk	basketballAmateur, basketballProfessional, wheelchairBasketball
Beverage Making	Making a lemonade, Mixing drinks	Mixing	makingCoffee, makingJuice, makingMilkTea, makingMixedDrinks, makingTea
Drill & Team Performance	Cheerleading, Baton twirling, Drum corps	BandMarching, MilitaryParade	chorus, marchingBand, parade, singingInKtv, singingOnStage
Oral & Facial Cleaning	Brushing teeth, Gargling mouthwash, Washing face, Washing hands	BrushingTeeth	brushingTeeth
Shaving & Body Care	Shaving, Shaving legs, Getting a tattoo, Getting a piercing	ShavingBeard	shavingBeard, tattooing
Surfing & Water Board Sports	Windsurfing, Surfing, Wakeboarding, Waterskiing	Surfing	kiteSurfing, surfing
Swimming	Swimming	BreastStroke, FrontCrawl	butterfly, swimmingAmateur, swimmingProfessional
Billiards & Board Games	Playing pool, Playing blackjack, Playing rubik cube, Rock-paper-scissors	Billiards	billiard, cardManipulation, playingChess, playingMahjong, solvingMagicCube
Hair Care	Blow-drying hair, Braiding hair, Brushing hair, Getting a haircut, Removing curlers	BlowDryHair, Haircut	hairCutting, hairstyleDesign
Ice & Snow Sports	Curling, Ice fishing, Shoveling snow, Snow tubing, Snowboarding, Skiing, Waxing skis	Skiing	iceSkating, makingSnowman, shovelingSnow, skiing, snowballFight
Makeup & Beauty Care	Applying sunscreen, Doing nails, Putting on makeup, Putting in contact lenses	ApplyEyeMakeup, ApplyLipstick	eyeMakeup, faceMassage, nailArtDesign, wearLipstick
Racket Sports	Playing racquetball, Playing squash, Playing badminton, Tennis serve with ball bouncing	TennisSwing	badminton, kicking-Shuttlecock, tennis, wheelchairTennis

Continued on next page

Table 1 – continued from previous page

Unified Category	ActivityNet	UCF101	FCVID
Pet Care & Dog Walking	Clipping cat claws, Grooming dog, Grooming horse, Bathing dog, Walking the dog	WalkingWithDog	cat, dog, walkingWithDog
Combat & Martial Arts	Capoeira, Doing karate, Doing a powerbomb, Sumo, Arm wrestling	SumoWrestling, Punch, Nunchucks	armWrestling, playing-WithNunChucks, street-Fighting, sumoWrestling, taekwondo
Gymnastics	Tumbling, Using parallel bars, Using the balance beam, Using the pommel horse, Using uneven bars, Slacklining	ParallelBars, BalanceBeam, PommelHorse, UnevenBars, StillRings, FloorGymnastics, TrampolineJumping	hulaHoop, parkour, rhythmicGymnastics, yoga
Wind Instrument Performance	Playing accordion, Playing bagpipes, Playing flauta, Playing harmonica, Playing saxophone	PlayingFlute, PlayingDaf, PlayingDhol	accordionPerformance, flutePerformance, harmonicaPerformance, saxophonePerformance, trumpetPerformance
Festivals & Celebrations	Decorating the Christmas tree, Carving jack-o-lanterns, Wrapping presents	BlowingCandles	birthday, decoratingChristmasTree, fireworksShow, marriageProposal, weddingCeremony, weddingReception
String Instrument Performance	Playing violin, Playing guitarra	PlayingViolin, PlayingGuitar, PlayingCello, PlayingSitar	celloPerformance, chamberMusic, guitarPerformance, symphony-OrchestraPerformance, violinPerformance
Outdoor Recreation & Entertainment	Building sandcastles, Kite flying, Hitting a pinata, Hopscotch, Fun sliding down, Riding bumper cars, Swinging at the playground, Starting a campfire	Swing, JugglingBalls, YoYo	amusementPark, bumperCars, flyingKites, penSpinning, picnic, pitchingATent, playground, yoyoTricks
Strength Training & Fitness	Clean and jerk, Snatch, Doing crunches, Doing step aerobics, Elliptical trainer	CleanAndJerks, BenchPress, BodyWeightSquats, Lunges, PullUps, PushUps, WallPushups, HandstandPushups, HandstandWalking, Jumping-Jack	barbellWorkout, dumbbellWorkout, pullUps, pushUps, sitUps, treadmill

Continued on next page

Table 1 – continued from previous page

Unified Category	ActivityNet	UCF101	FCVID
Cleaning & Housework	Cleaning shoes, Cleaning sink, Cleaning windows, Hand washing clothes, Ironing clothes, Mooping floor, Vacuuming floor, Washing dishes, Polishing furniture, Polishing shoes	MoppingFloor	carWashing, cleaningAppliance, cleaningCarpet, cleaningFloor, cleaningWindows, mowing, washingAnInfant, washingDishes
Knitting & Handicrafts	Knitting	Knitting	knitting, makingBookmark, makingBracelets, makingCeramicCraft, makingEarrings, makingFestivalCards, makingPaperFlowers, makingPaperPlane, makingPencilCases, makingPhoneCases, makingPhotoFrame, makingRings, makingShorts, makingWallet, paperCutting, sculpting
Cooking & Food Preparation	Making a cake, Making a sandwich, Making an omelette, Preparing pasta, Preparing salad, Baking cookies, Peeling potatoes, Sharpening knives	PizzaTossing, CuttingInKitchen	Cutting, makingCake, makingChineseDumplings, makingCookies, makingEggTarts, makingFrenchFries, makingHotdog, makingIcecream, makingPizza, makingSalad, makingSandwich, makingSushi, roastingTurkey, deliciousFood, diningAtRestaurant, dinnerAtHome, groupBanquet

2 Temporal Generator Details

In this section we provide a comprehensive description of the temporal generator G_θ introduced in the main text (VidInc-Synthetic). We detail its architecture, training objectives, optimization procedure, hyperparameters, and model configuration.

2.1 Architecture of G_θ

The generator operates on frame-level feature sequences. Given a normalized video clip $\mathbf{X} \in \mathbb{R}^{T \times D}$ with $T = 25$ frames and $D = 768$ (CLIP ViT-L/14 features), the forward pass for a target domain $k \in \{1, \dots, K_n\}$ ($K_n = 3$) is defined by three successive operations:

$$\mathbf{H} = \mathbf{X} \mathbf{W}_{\text{proj}} \in \mathbb{R}^{T \times H}, \quad (1)$$

$$\mathbf{Z}^{(k)} = \text{Transformer}([\mathbf{e}_k; \mathbf{H}]) \in \mathbb{R}^{(T+1) \times H}, \quad (2)$$

$$\tilde{\mathbf{X}}^{(k)} = \mathbf{X} + \mathbf{Z}_{1:T,:}^{(k)} \mathbf{W}_{\text{out}}, \quad (3)$$

where $H = 768$ is the latent dimensionality. The domain index k is the only conditioning input.

- **Latent Projection (1):** The linear layer $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{D \times H}$ maps each frame independently into the latent space.
- **Domain-Conditioned Encoding (2):** A learnable domain embedding $\mathbf{e}_k = \mathbf{E}[k] \in \mathbb{R}^H$ is taken from an embedding table $\mathbf{E} \in \mathbb{R}^{(1+K_n) \times H}$. The table is initialized with orthogonal rows scaled by $\sqrt{H}$ to ensure diverse, well-conditioned starting vectors. The embedded token is prepended to the frame sequence $\mathbf{H}$, and the resulting $(T+1)$ tokens are processed by a 2-layer Transformer encoder with Pre-Layer Normalization (Pre-LN) and $H = 768$, 4 attention heads. The Transformer captures long-range inter-frame dependencies under the influence of the domain token.
- **Residual Synthesis (3):** After the Transformer, the output at the domain-token position (first row of $\mathbf{Z}^{(k)}$) is discarded. Only the T frame-token outputs are projected back to the original feature dimension via $\mathbf{W}_{\text{out}} \in \mathbb{R}^{H \times D}$, and a residual connection adds the result to the input $\mathbf{X}$. The output projection weights are initialized to zero, ensuring that at the beginning of training the generator acts as identity and the training signal comes solely from the losses.

This formulation guarantees that G_θ learns an additive shift on the temporal manifold, preserving the underlying semantic skeleton of the original video.

2.2 Optimization Objectives

Training the generator involves a multi-objective functional together with an auxiliary classifier F_ψ that assesses the quality of the generated features. In the following, $\mu(\cdot)$ denotes temporal mean pooling, producing a feature vector of dimension D for each video.

2.2.1 Generator Losses

1. **Distributional Divergence ($\mathcal{L}_{\text{nov}}, \mathcal{L}_{\text{div}}$).** Both terms rely on the Gromov–Wasserstein distance (GWD) computed between video-level mean features. Given two sets of features $\mathbf{A}_1, \mathbf{A}_2$ and $\mathbf{B}_1, \mathbf{B}_2$, we define the Gromov inner product distance

$$\text{GED}(\mathbf{A}_1, \mathbf{A}_2; \mathbf{B}_1, \mathbf{B}_2) = 2 \text{OT}(\mathbf{A}_1, \mathbf{B}_1) - \text{OT}(\mathbf{A}_1, \mathbf{A}_2) - \text{OT}(\mathbf{B}_1, \mathbf{B}_2), \quad (4)$$

where $\text{OT}(\cdot, \cdot)$ is the entropic optimal transport distance (Sinkhorn algorithm) with regularization $\varepsilon = 0.5$. For a batch of size B , we split it into two halves of size $n = B/2$. Then, for each novel domain k , the novelty loss is

$$\mathcal{L}_{\text{nov}} = \sum_{k=1}^{K_n} \text{GED}(\mu(\tilde{\mathbf{X}}_1^{(k)}), \mu(\tilde{\mathbf{X}}_2^{(k)}); \mu(\mathbf{X}_1), \mu(\mathbf{X}_2)), \quad (5)$$

which encourages the synthetic domain distributions to move away from the source manifold. The diversity loss for pairs of synthetic domains is

$$\mathcal{L}_{\text{div}} = \sum_{1 \leq i < j \leq K_n} \text{GED}(\mu(\tilde{\mathbf{X}}_1^{(i)}), \mu(\tilde{\mathbf{X}}_2^{(i)}); \mu(\tilde{\mathbf{X}}_1^{(j)}), \mu(\tilde{\mathbf{X}}_2^{(j)})), \quad (6)$$

forcing the domains to occupy distinct regions of the feature space.

2. **Structural Integrity** ($\mathcal{L}_{\text{cyc}}$). Cycle-consistency is enforced by feeding the synthetic features back into the same G_θ conditioned on the source domain (domain index 0) and requiring reconstruction of the original features:

$$\mathcal{L}_{\text{cyc}} = \sum_{k=1}^{K_n} \|G_\theta(\tilde{\mathbf{X}}^{(k)}, 0) - \mathbf{X}\|_1, \quad (7)$$

where the generator input $\tilde{\mathbf{X}}^{(k)}$ is detached from the computation graph to avoid back-propagation through a nested generation.

3. **Semantic Anchor** ($\mathcal{L}_{\text{ce}}$). A frozen pre-trained classifier ϕ (architecture identical to F_ψ , see below) ensures class-consistency:

$$\mathcal{L}_{\text{ce}} = \sum_{k=1}^{K_n} \text{CE}(\phi(\mu(\tilde{\mathbf{X}}^{(k)})), \mathbf{y}), \quad (8)$$

where $\mathbf{y}$ are the ground-truth labels of the source batch $\mathbf{X}$ and CE denotes the cross-entropy loss.

The total generator loss is a weighted combination:

$$\mathcal{L}_G = -\lambda_d(\mathcal{L}_{\text{nov}} + \mathcal{L}_{\text{div}}) + \lambda_c \mathcal{L}_{\text{cyc}} + \lambda_{\text{ce}} \mathcal{L}_{\text{ce}}, \quad (9)$$

with $\lambda_d = 0.5$, $\lambda_c = 0.01$, and $\lambda_{\text{ce}} = 0.1$.

2.2.2 Auxiliary Classifier (F_ψ) and Quality Assessment

An auxiliary classifier F_ψ (a small MLP, see Table 3) is jointly trained to evaluate the fidelity of the generated features. During the update of F_ψ , the generator G_θ is frozen; the loss is a mixture of cross-entropy on real and synthetic mean features:

$$\mathcal{L}_F = (1 - \alpha) \text{CE}(F_\psi(\mu(\mathbf{X})), \mathbf{y}) + \frac{\alpha}{K_n} \sum_{k=1}^{K_n} \text{CE}(F_\psi(\mu(\tilde{\mathbf{X}}^{(k)})), \mathbf{y}), \quad (10)$$

with mixing coefficient $\alpha = 0.5$.

Importantly, F_ψ does not provide gradients to G_θ during the generator update. Instead, its loss $\mathcal{L}_F$ is monitored and used to select the best checkpoint of the generator (G_θ and F_ψ are saved together). This ensures that only generators producing features that are discriminatively consistent with the source distribution are retained.

2.3 Training Procedure

The training proceeds in epochs, each consisting of $I = 10$ iterations. Within one iteration, we first update F_ψ using a batch drawn from the full dataset, and then update G_θ by sequentially processing class-balanced mini-batches. The detailed algorithm is given as Algorithm 1.

For the generator update, we iterate over all classes that have at least $k_{\text{shot}} = 32$ samples. For each such class, a batch of size $B = 32$ is drawn; the batch is halved to compute the OT-based losses. The total generator loss is computed, gradients are accumulated, and after all classes have been processed, the accumulated gradients are clipped and G_θ is updated. This class-wise accumulation ensures a balanced contribution from every class despite the uneven dataset sizes.

The semantic classifier ϕ is pre-trained for 15 epochs on the mean features of the source dataset and then frozen. All optimizers are Adam: learning rate 10^{-4} for G_θ and 5×10^{-4} for F_ψ , with weight decay 10^{-5} . Training runs for $E = 25$ epochs.

2.4 Hyperparameters

Table 2 summarizes the key hyperparameters used for training the generator on ActivityNet.

Algorithm 1 Training of the Temporal Generator G_θ

Require: Dataset $\mathcal{D}$ of frame sequences $\{(\mathbf{X}_i, y_i)\}$; number of novel domains K_n ; frozen classifier ϕ ; hyperparameters $\lambda_d, \lambda_c, \lambda_{ce}, \alpha, \eta_G, \eta_F$, epochs E , iterations per epoch I , minimum samples per class k_{shot} .

Initialize: Initialize G_θ and F_ψ ; let $\mu(\cdot)$ denote temporal mean pooling.

```
1: Pretrain  $\phi$  on  $\mu(\mathbf{X})$  for 15 epochs; freeze  $\phi$ .
2: for epoch  $e = 1$  to  $E$  do
3:   for step  $t = 1$  to  $I$  do
4:     // Stage 1: update  $F_\psi$  (with  $G_\theta$  frozen)
5:     Sample a batch  $\mathcal{B}_F = \{(\mathbf{X}, \mathbf{y})\}$  from  $\mathcal{D}$ ; for each  $k = 1, \dots, K_n$ , generate  $\tilde{\mathbf{X}}^{(k)} = G_\theta(\mathbf{X}, k)$  with no gradient.
6:     Compute  $\mathcal{L}_F$  according to Eq. (10); update  $F_\psi$  using its optimizer.
7:     // Stage 2: update  $G_\theta$  (with  $F_\psi$  frozen)
8:      $\mathbf{g} \leftarrow \mathbf{0}$ ; let  $\mathcal{C}_{\text{valid}}$  be the set of classes with at least  $k_{\text{shot}}$  samples.
9:     for each class  $c \in \mathcal{C}_{\text{valid}}$  do
10:      Sample a batch  $\mathcal{B}_c = \{(\mathbf{X}, c)\}$  of size  $B$ ; split into two halves  $\mathbf{X}_1, \mathbf{X}_2$ .
11:      For each  $k = 1, \dots, K_n$ , generate  $\tilde{\mathbf{X}}^{(k)} = G_\theta(\mathbf{X}, k)$ .
12:      Compute  $\mathcal{L}_{\text{nov}}$  (Eq. (5)),  $\mathcal{L}_{\text{div}}$  (Eq. (6)),  $\mathcal{L}_{\text{cyc}}$  (Eq. (7)),  $\mathcal{L}_{ce}$  (Eq. (8)).
13:      Compute total generator loss  $\mathcal{L}_G = -\lambda_d(\mathcal{L}_{\text{nov}} + \mathcal{L}_{\text{div}}) + \lambda_c\mathcal{L}_{\text{cyc}} + \lambda_{ce}\mathcal{L}_{ce}$ .
14:      Accumulate gradient:  $\mathbf{g} \leftarrow \mathbf{g} + \frac{1}{|\mathcal{C}_{\text{valid}}|} \nabla_{\theta} \mathcal{L}_c$ .
15:     end for
16:     Clip gradients; update  $G_\theta$ :  $\theta \leftarrow \theta - \eta_G \mathbf{g}$ .
17:   end for
18: end for
19: return  $G_\theta$  (and  $F_\psi$  for checkpoint selection)
```

2.5 Model Configuration

Table 3 lists the exact specifications of all networks involved.

2.6 Data Pre-processing

All video features are extracted with CLIP ViT-L/14, taking $T = 25$ uniformly sampled frames per video. Each frame feature is L2-normalized before being fed to the generator. During training, the full dataset is used to train F_ψ , while G_θ is trained with k_{shot} -shot batches drawn per class to maintain balanced gradient signals across the 200 classes.

3 Experimental Setups

Evaluation Metrics. Following [1], we adopt **Average Accuracy** $\text{ACC} = \frac{1}{T} \sum_{i=1}^T A_{T,i}$ and **Backward Transfer** $\text{BWT} = \frac{1}{T-1} \sum_{i=1}^{T-1} (A_{T,i} - A_{i,i})$, where T is the total number of sequential tasks, as evaluation metrics. BWT serves as a critical proxy for quantifying catastrophic forgetting: negative values indicate performance decay on earlier tasks. Additionally, **Maximum Mean Discrepancy (MMD)** [2] with a Gaussian RBF kernel quantifies the magnitude of distribution shifts between domains.

Representative Baselines. We evaluate three paradigms: (1) Regularization-based: EWC [3], SI [4]. (2) Replay-based: ER [5], GDumb [6], DER++ [7]. (3) **Fine-tuning**, which trains sequentially on each task without access to previous data and without any forgetting mitigation; and **Joint Training**, which trains over the entire data stream simultaneously.

Implementation Details. CLIP ViT-L/14 [8] extracts 768-dimensional features from 25 uniformly sampled frames per video, following the observation that self-supervised pre-trained representations provide a strong prior for incremental learning [9]. The baseline architecture consists of a 2-layer Transformer encoder (hidden dimension 768, 8 attention heads) followed by a linear classification head. Optimization uses Adam [10] ($lr = 10^{-4}$, batch size 32), and replay methods employ a fixed buffer of 2,000 samples.

Table 2: Training hyperparameters for the VidInc-Synthetic generator.

Parameter	Value
Generator learning rate η_G	10^{-4}
Auxiliary classifier learning rate η_F	5×10^{-4}
Weight decay (both)	10^{-5}
Batch size B (both stages)	32
k_{shot} per class	32
Epochs E	25
Iterations per epoch I	10
Pre-training epochs for ϕ	15
K_n (number of novel domains)	3
Latent dimension H	768
Transformer layers	2
Transformer attention heads	4
λ_d	0.5
λ_c	0.01
λ_{ce}	0.1
Sinkhorn regularization ε	0.5
Mixing weight α (Eq. 10)	0.5
Optimizer	Adam

Protocol Specifics per Benchmark.

- **VidInc-Class:** Models are trained on sequential tasks $\mathcal{T}_1, \dots, \mathcal{T}_{10}$ using a fixed stochastic seed (42) for task partitioning. Final performance is quantified by Average Accuracy (ACC) and Backward Transfer (BWT) after the full stream integration.
- **VidInc-CrossDomain:** Models encounter three domain-specific tasks sequentially, each covering the full 43-category label space. To account for sensitivity to domain ordering, we report the mean and standard deviation across all six possible domain permutations.
- **VidInc-Synthetic:** Models follow a fixed curriculum (Source $\rightarrow$ syn1 $\rightarrow$ syn2 $\rightarrow$ syn3) consisting of four tasks within a 200-category space. ACC and BWT are reported to measure adaptation capacity and the magnitude of cross-domain interference.

Incremental tasks are optimized for 15–20 epochs to achieve peak performance, whereas the **Joint Training** baseline utilizes 100 epochs to guarantee full convergence on the aggregated corpus.

4 Per-Order Breakdown for VidInc-CrossDomain

Table 4 provides the per-order ACC and BWT for all methods.

Table 3: Architectural details of the temporal generator and auxiliary classifiers.

Module	Layer / Component	Specification
G_θ	Input	$\mathbf{X} \in \mathbb{R}^{25 \times 768}$, L2-normalized
	Frame Projection	Linear(768, 768)
	Domain Embedding $\mathbf{E}$	Embedding(4, 768), orthogonal init, scaled by $\sqrt{768}$
	Transformer Encoder	2 layers, $d_{\text{model}} = 768$, 4 heads, Pre-LN, no dropout
	Output Projection $\mathbf{W}_{\text{out}}$	Linear(768, 768), zero-initialized
F_ψ and ϕ (identical)	MLP	Linear(768, 128), no bias
		LayerNorm(128, no bias)
		GELU activation
		Linear(128, 200), no bias

Table 4: Per-Order ACC and BWT (%) on VidInc-CrossDomain. A: ActivityNet, F: FCVID, U: UCF101.

Method	Order						Mean $\pm$ Std
	A $\rightarrow$ U $\rightarrow$ F	A $\rightarrow$ F $\rightarrow$ U	U $\rightarrow$ A $\rightarrow$ F	U $\rightarrow$ F $\rightarrow$ A	F $\rightarrow$ A $\rightarrow$ U	F $\rightarrow$ U $\rightarrow$ A	
ACC							
Fine-tuning	42.86	56.61	33.43	49.16	54.20	65.09	50.22 $\pm$ 11.09
EWC	58.44	58.09	47.63	57.41	65.13	66.22	58.82 $\pm$ 6.67
SI	28.84	13.59	24.43	6.52	33.14	29.86	22.73 $\pm$ 10.45
ER	67.30	78.09	66.92	73.82	86.13	84.92	76.20 $\pm$ 8.36
GDumb	73.28	77.46	77.60	75.67	76.62	74.69	75.89 $\pm$ 1.69
DER++	84.13	84.09	86.35	90.58	91.35	91.36	87.98 $\pm$ 3.52
BWT							
Fine-tuning	-72.04	-54.08	-86.55	-64.07	-59.07	-42.98	-63.12 $\pm$ 16.42
EWC	-46.93	-46.67	-59.30	-47.56	-38.75	-35.79	-45.83 $\pm$ 8.34
SI	-84.16	-85.64	-87.17	-93.16	-87.74	-91.80	-88.28 $\pm$ 3.46
ER	-36.25	-19.00	-33.32	-27.41	-10.09	-12.43	-23.08 $\pm$ 11.83
GDumb	10.91	13.28	4.41	1.44	33.59	10.56	9.45 $\pm$ 12.03
DER++	-7.38	-9.08	-7.74	-2.35	-1.69	-1.12	-3.45 $\pm$ 3.71

References

- [1] D. Lopez-Paz and M. A. Ranzato, “Gradient episodic memory for continual learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, vol. 30.
- [2] A. Gretton et al., “A kernel two-sample test,” *Journal of Machine Learning Research*, vol. 13, no. 1, pp. 723–773, 2012.
- [3] J. Kirkpatrick et al., “Overcoming catastrophic forgetting in neural networks,” *Proceedings of the National Academy of Sciences*, vol. 114, no. 13, pp. 3521–3526, 2017.
- [4] F. Zenke, B. Poole, and S. Ganguli, “Continual learning through synaptic intelligence,” in *Proceedings of the International Conference on Machine Learning (ICML)*. PMLR, 2017, pp. 3987–3995.
- [5] P. Weber, D. Wälchli, M. Zeqiri, and P. Koumoutsakos, “Remember and forget experience replay for multi-agent reinforcement learning,” *arXiv preprint arXiv:2203.12345*, 2022.

- [6] A. Prabhu, P. H. S. Torr, and P. K. Dokania, “Gdumb: A simple approach that questions our progress in continual learning,” in *Proceedings of the European Conference on Computer Vision (ECCV)*. Springer, 2020, pp. 524–540.
- [7] P. Buzzega et al., “Dark experience for general continual learning: a strong, simple baseline,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020, vol. 33, pp. 15920–15930.
- [8] A. Radford et al., “Learning transferable visual models from natural language supervision,” in *Proceedings of the International Conference on Machine Learning (ICML)*. PMLR, 2021, pp. 8748–8763.
- [9] E. Fini et al., “Self-supervised models are continual learners,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2022, pp. 9621–9630.
- [10] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *arXiv preprint arXiv:1412.6980*, 2014.